

Day 8: Simple Menu Choice

1. Day Number

Day 8

2. Topic Name

Creating a menu using **if-elif-else**

3. Connection with Previous Learning

Last week, you created simple banking actions such as:

- Opening a bank account
- Depositing money
- Withdrawing money
- Checking different conditions

Today, you will create a **menu** so the user can choose which banking action they want to perform.

For example:

```
1. Check Balance
2. Exit
```

The program will read the user's choice and decide what message to display.

4. Important Topics

Menu

A menu displays a list of available options.

```
1. Check Balance
2. Exit
```

input()

`input()` receives a value from the user.

```
choice = input("Enter your choice: ")
```

Remember that `input()` always returns a **string**.

Integer conversion

You can use `int()` to convert numeric text into an integer.

```
choice = int(input("Enter your choice: "))
```

If the user enters "1", Python converts it into the integer 1.

if-elif-else

These statements allow the program to select one action from multiple possibilities.

```
if condition_one:
    # First action
elif condition_two:
    # Second action
else:
    # Invalid or unexpected action
```

5. Foundational Notes

What is a menu-driven program?

A menu-driven program shows multiple options and asks the user to select one.

```
1. View profile
2. Logout
```

The program then checks the selected option.

Why convert the choice to an integer?

When the user enters:

```
1
```

Python initially receives it as:

```
"1"
```

This is a string.

After using `int()`, it becomes:

```
1
```

This is an integer.

Therefore, this comparison works:

```
choice == 1
```

How does if-elif-else work?

Python checks conditions from top to bottom:

1. It checks the `if` condition.
2. If that is false, it checks the `elif` condition.
3. If all conditions are false, it runs the `else` block.

Only **one matching block** is executed.

Indentation is important

The code belonging to a condition must be indented.

```
if condition:
    print("This line belongs to the if block")
```

6. Easy Example

Imagine a small application with this menu:

- ```
1. Say Hello
2. Say Goodbye
```

A possible example is:

```
choice = int(input("Enter your choice: "))

if choice == 1:
 print("Hello!")
elif choice == 2:
 print("Goodbye!")
else:
 print("Invalid choice")
```

This example demonstrates how one user choice can produce different outputs.

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## 7. Problem Statement

Create a simple banking menu.

The program should display:

- ```
1. Check Balance
2. Exit
```

Ask the user to enter a choice.

Then:

- If the user enters 1, print a balance-related message.
- If the user enters 2, print an exit message.
- For any other number, print an invalid-choice message.

Do not build actual deposit or withdrawal functionality today. The focus is only on understanding **menu selection**.

8. Concepts Used

You will use:

- Variables
- Strings

- Integers
 - `print()`
 - `input()`
 - `int()`
 - Comparison operator `==`
 - `if`
 - `elif`
 - `else`
 - Functions
 - Function calls
-

9. Thought Process

Think about the problem in this order:

Step 1: Display the menu

Show the two options to the user.

```
1. Check Balance
2. Exit
```

Step 2: Ask for the choice

Store the user's selected option in a variable.

Possible variable name:

```
choice
```

Step 3: Convert the input

Because `input()` gives you a string, convert it to an integer using `int()`.

Step 4: Check for option 1

If the choice is 1, display the balance message.

Step 5: Check for option 2

If the choice is 2, display the exit message.

Step 6: Handle invalid choices

If the user enters another number, such as 3 or 10, display:

```
Invalid choice
```

Step 7: Organize the logic inside a function

Place the menu logic inside a function and call that function to run the program.

10. Pseudocode

```
START

DEFINE a function for the banking menu

    DISPLAY "1. Check Balance"
    DISPLAY "2. Exit"

    ASK the user to enter a choice
    CONVERT the choice into an integer

    IF choice is equal to 1
        DISPLAY the balance message

    ELSE IF choice is equal to 2
        DISPLAY the exit message

    ELSE
        DISPLAY "Invalid choice"

CALL the banking menu function

END
```

11. Suggested Solving Approach: Functional Approach

Use a function to keep the program organized.

Suggested structure:

```
main program
├── banking_menu()
│   ├── display options
│   ├── receive user choice
│   ├── check choice
│   └── display result
```

You can create one function such as:

```
def banking_menu():
```

Inside the function:

1. Print the menu.
2. Ask for the choice.
3. Convert the choice to an integer.
4. Use if-elif-else.
5. Print the appropriate message.

Finally, call the function.

12. Easy Edge Cases

Edge case 1: Invalid numeric choice

Input:

5

Since option 5 does not exist, the program should print:

Invalid choice

Edge case 2: Choice 0

Input:

0

Since 0 is not a valid option, print:

Invalid choice

Edge case 3: Text instead of a number

Input:

hello

Python cannot convert "hello" into an integer using `int()`.

It will produce a `ValueError`.

For today's basic version, assume that the user normally enters a number. As an optional improvement, you can later use `try-except` to handle text input safely.

Edge case 4: Numeric text

Input:

1

Although `input()` initially receives "1" as a string, `int()` converts it into the integer 1.

13. Expected Input

Example 1

Enter your choice: 1

Example 2

Enter your choice: 2

Example 3

```
Enter your choice: 7
```

14. Expected Output

When the user selects option 1

```
1. Check Balance
2. Exit
Enter your choice: 1
Your balance is available.
```

When the user selects option 2

```
1. Check Balance
2. Exit
Enter your choice: 2
Thank you. Exiting the program.
```

When the user enters an invalid number

```
1. Check Balance
2. Exit
Enter your choice: 7
Invalid choice.
```

15. Hint Only

Create a function that prints the two menu options. Store the converted user input in a variable named choice.

Then think about these three conditions:

```
Is choice equal to 1?
Is choice equal to 2?
Is it something else?
```

Use `if` for the first choice, `elif` for the second choice, and `else` for invalid choices.